

Gene-Level Readouts and the Metabolic Phenotype

What a Deletion's Transcriptome or Proteome Says About Its Amino-Acid Pools

and Pigment Titers, Panel by Panel

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Abstract

A deletion's gene-level readout carries its metabolic phenotype, and how much of it depends on which readout. On the deletions the panels share, a ridge from the Messner 2023 proteome to the 19 Mulleder 2016 amino-acid pools reads a median out-of-fold Pearson of 0.41 over 4,400 deletions, every amino acid above its permutation null, proline at 0.61 through the prolyl-tRNA synthetase and alanine at 0.55 through *ALT1*; the same ridge from the Kemmeren 2014 transcriptome reads 0.18 over 1,416 deletions, and from the Nadal-Ribelles 2025 perturb-seq pseudobulk 0.10 over 2,113. The single features behind the strong reads are the pathway's own enzymes and synthetases, with the sign a pool-and-flux picture predicts. On the betaxanthin titer proxy of Cachera 2023 the three sources read alike, 0.43, 0.35 and 0.41, on a respiratory signature; the beta-carotene colony score reads 0.07 to 0.16. Alike transcriptomes have alike amino-acid profiles (Mantel Spearman -0.20 over a million Kemmeren pairs against a null of 0.03), while the proteome, which predicts the individual amino acids best, has the weakest whole-profile read (-0.05).

Nadal-Ribelles does not replicate itself across batches and barely agrees with either other source on the same deletions, yet it reads the betaxanthin phenotype as well as they do and half of the transcriptome's amino-acid signal. Agreement with an earlier compendium is therefore not the test of whether a readout is worth training on; the metabolic read is an independent one, and on it the perturb-seq atlas carries a large-effect program and little of the smaller ones. Zelezniak 2018 and da Silveira 2014 share under 125 deletions with any source and read null, as their size predicts. Two things are not measured: whether a shared plate structure between the Mulleder and Messner panels contributes to the proteome's read, and whether the encoder trained on the proteome carries the amino acids in its cell vector.

How to read this. section 1 is the question and the panels that can answer it. section 2 is the answer, read by read. section 3 is the statistics, and section 4 is what the evidence leaves open.

Working note: `notes/experiments.028-knockout-expression.metabolic-phenotype-correlation.md`

Generating script: `experiments/028-knockout-expression/scripts/expression_metabolic_yko_correlation.py`

Results: `experiments/028-knockout-expression/results/expression_metabolic_yko_correlation.json`

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Contents

1	The question, and the panels that can answer it ✗	2
2	What the reads say ✗	4
2.1	The proteome carries the amino-acid pools; the transcriptome carries less of them ✗	4
2.2	The pigment screens read the same way ✗	5
2.3	Alike transcriptomes, alike amino-acid profiles ✗	5
2.4	Nadal-Ribelles reads the phenotype through one program ✗	5
2.5	The small panels read null, as their size predicts ✗	6
3	The statistics ✗	6
3.1	Matrices ✗	6
3.2	Three reads ✗	6
4	What the evidence leaves open ✗	7

1 The question, and the panels that can answer it ✗

A **gene-level readout** is a profile with one number per gene: a transcriptome or a proteome. A **metabolic phenotype** is a small vector of metabolite readouts or a single titer. The question is whether the first carries the second: given only a deletion's transcriptome or proteome, can its amino-acid pools or its pigment titer be read off? If yes, a model that predicts the gene-level readout has already learned most of what a metabolic head needs, and the two heads share an encoder for a reason. If no, the metabolic head must find its signal elsewhere.

The knockout collection is the bridge. Three sources profile single-gene deletions of the same collection: the Kemmeren 2014 microarray compendium ([doi:10.1016/j.cell.2014.02.054](https://doi.org/10.1016/j.cell.2014.02.054); 1,484 deletions, mRNA $\log_2$ ratio to wild type over 6,169 reporters), the Messner 2023 knockout proteome ([doi:10.1016/j.cels.2022.12.003](https://doi.org/10.1016/j.cels.2022.12.003); 4,549 deletions after averaging duplicate ORFs, $\log_2$ protein quantity over the HIS3 reference across 1,830 proteins), and the Nadal-Ribelles 2025 perturb-seq atlas ([doi:10.1101/2025.04.30.651486](https://doi.org/10.1101/2025.04.30.651486); 2,243 deletions with at least 50 cells, pseudobulk $\log_2$ fold change over 5,404 genes). Six metabolic panels score deletions of the same collection: Mulleder 2016 ([doi:10.1016/j.cell.2016.09.007](https://doi.org/10.1016/j.cell.2016.09.007); 19 intracellular amino acids in mM, 4,678 deletions), Cooper 2010 ([doi:10.1101/gr.105825.110](https://doi.org/10.1101/gr.105825.110); 16 capillary-electrophoresis peaks as a ratio to the plate mean, 4,313 deletions), Cachera 2023 ([doi:10.1038/s41467-023-42394-3](https://doi.org/10.1038/s41467-023-42394-3); betaxanthin corrected fluorescence, 4,719 deletions), Ozaydin 2013 ([doi:10.1016/j.ymben.2012.07.010](https://doi.org/10.1016/j.ymben.2012.07.010); beta-carotene colony color score from -5 to $+5$, 4,474 deletions), Zelezniak 2018 ([doi:10.1016/j.cels.2018.08.001](https://doi.org/10.1016/j.cels.2018.08.001); central-carbon metabolites in 95 kinase deletions), and da Silveira 2014 ([doi:10.1016/j.molcel.2014.11.014](https://doi.org/10.1016/j.molcel.2014.11.014); 135 lipids in 127 deletions). SRC: [experiments/028-knockout-expression/results/expression_metabolic_yko_correlation.json](#)

Every (panel, source) pair, with the deletions the two share, is in table 1. The three large amino-acid and pigment panels share 1,300 to 1,450 deletions with Kemmeren and 3,900 to 4,400 with Messner. Zelezniak and da Silveira share under 125 deletions with any source, too few for a ridge over thousands of features, and their reads are carried in the table as the null they are.

Why the three sources are not interchangeable. The three profile the same deletions and barely agree with one another. Messner protein against Kemmeren mRNA for the same deletion reads a per-deletion median Pearson of 0.04 and a per-gene median of 0.08; Messner against Nadal-Ribelles reads 0.01 on both, and Nadal-Ribelles does not replicate itself across its own batches (cross-batch r of 0.04). So the metabolic read is a second, independent test of each source: a source that carries the metabolic phenotype carries something real whether or not it agrees with the others, and a source that carries nothing has no such defense. SRC: [experiments/028-knockout-expression/results/proteome_expression_covariation.json](#)

Table 1: Every (metabolic panel, molecular source) read on the deletions the two share. Shared is the number of deletions profiled on both sides; genes is the number of features after the coverage filter; scored is the number of metabolites with at least 40 deletions. The ridge score is the out-of-fold Pearson of a ridge from the whole profile to the metabolite, its median over scored metabolites, and the count above the permutation 95th percentile of the same statistic. Mantel is the Spearman between strain-pair profile similarity and strain-pair metabolite distance, with the permutation 95th percentile of $|\rho|$ in parentheses; it needs at least three metabolites and 40 complete deletions.

SRC: `experiments/028-knockout-expression/scripts/expression_metabolic_yko_correlation.py`

panel		source	shared	genes	scored	median ridge	r	above null	best metabolite	Mantel ρ (null)
Mulleder amino acids	2016	Kemmeren mRNA	1,416	6,169	19	0.18	18 of 19	Ala (0.31)	-0.196 (0.033)	
Mulleder amino acids	2016	Messner protein	4,400	1,536	19	0.41	19 of 19	Pro (0.61)	-0.046 (0.003)	
Mulleder amino acids	2016	Nadal-Ribelles mRNA	2,113	4,412	19	0.10	17 of 19	Ser (0.20)	-0.067 (0.034)	
Cooper 2010 amino acids	2010	Kemmeren mRNA	1,317	6,169	17	0.10	12 of 17	lysine b (0.39)	-0.150 (0.077)	
Cooper 2010 amino acids	2010	Messner protein	3,912	1,526	17	0.08	14 of 17	lysine b (0.30)	+0.002 (0.010)	
Cooper 2010 amino acids	2010	Nadal-Ribelles mRNA	1,957	4,412	17	0.01	2 of 17	lysine b (0.10)	-0.059 (0.076)	
Cachera 2023 betaxanthin	2023	Kemmeren mRNA	1,439	6,169	1	0.35	1 of 1	betaxanthin (0.35)	-	
Cachera 2023 betaxanthin	2023	Messner protein	4,308	1,530	1	0.43	1 of 1	betaxanthin (0.43)	-	
Cachera 2023 betaxanthin	2023	Nadal-Ribelles mRNA	2,153	4,412	1	0.41	1 of 1	betaxanthin (0.41)	-	
Ozaydin 2013 betacarotene	2013	Kemmeren mRNA	1,361	6,169	1	0.12	1 of 1	visual score (0.12)	-	
Ozaydin 2013 betacarotene	2013	Messner protein	4,128	1,535	1	0.16	1 of 1	visual score (0.16)	-	
Ozaydin 2013 betacarotene	2013	Nadal-Ribelles mRNA	1,963	4,412	1	0.07	1 of 1	visual score (0.07)	-	
Zelezniak metabolome	2018	Kemmeren mRNA	93	6,169	17	-0.05	0 of 17	e4p (0.22)	-	
Zelezniak metabolome	2018	Messner protein	88	1,522	17	-0.05	1 of 17	xu5p-D;ru5p-D (0.20)	-	
Zelezniak metabolome	2018	Nadal-Ribelles mRNA	47	4,421	13	0.02	2 of 13	r5p (0.11)	-	
da Silveira lipids	2014	Kemmeren mRNA	121	6,169	147	-0.00	23 of 147	LysoPI18:0 (0.40)	-	
da Silveira lipids	2014	Messner protein	116	1,502	147	0.04	34 of 147	PE36:1 (0.28)	-	
da Silveira lipids	2014	Nadal-Ribelles mRNA	65	4,420	141	-0.15	9 of 141	PC44:2 (0.26)	-	

2 What the reads say ✗

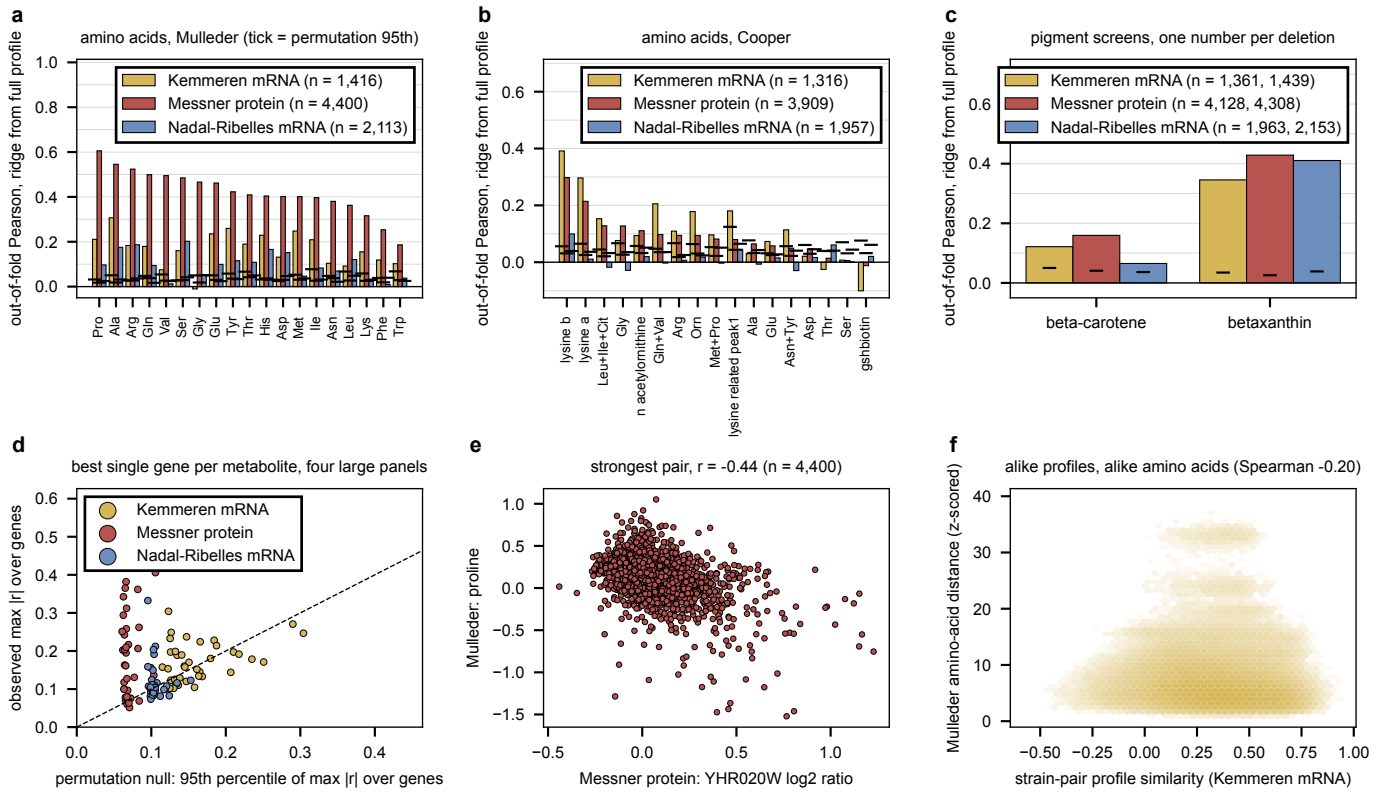


Figure 1: A deletion's proteome predicts its amino-acid pools at 0.4 to 0.6 out of fold, its transcriptome at 0.1 to 0.3, and every bar of the two large amino-acid panels clears its permutation null. a, b, out-of-fold Pearson of a ridge from the whole profile to each amino acid (Muller, Cooper), one bar per source, the black tick the permutation 95th percentile of the same statistic. c, the two pigment screens. d, the largest single-feature $|r|$ per metabolite against the permutation null of that maximum, for the four large panels; the Kemmeren null is wide because the metabolite values are heavy tailed. e, the strongest single pair, the prolyl-tRNA synthetase protein against the proline pool. f, profile similarity of a strain pair against the amino-acid distance of the same pair over 1.0 million Kemmeren pairs. Generated by `expression_metabolic_yko_correlation.py`.

2.1 The proteome carries the amino-acid pools; the transcriptome carries less of them ✗

On the 19 Muller amino acids the ridge from the Messner proteome reads a median out-of-fold Pearson of 0.41 over 4,400 shared deletions, every one of the 19 above its permutation 95th percentile (which sits between 0.03 and 0.08); proline reads 0.61, alanine 0.55, arginine 0.52, glutamine 0.50. The same ridge from the Kemmeren transcriptome reads a median of 0.18 over 1,416 shared deletions, 18 of 19 above null; alanine reads 0.31, tyrosine 0.26, methionine 0.25 (table 2). Cooper 2010 repeats the ordering at lower strength, a median of 0.08 from protein and 0.10 from mRNA, with the lysine peaks at 0.30 to 0.39; its values are ratios to a plate mean from a single measurement, so its ceiling is lower than Muller's before any source is asked to reach it. SRC: `experiments/028-knockout-expression/results/expression_metabolic_yko_correlation.json`

The single features behind the strong reads are pathway enzymes. Proline's top protein is the prolyl-tRNA synthetase YHR020W ($r = -0.44$ over 4,400 deletions), then *ARO2* and *THR1* at -0.41 . Alanine's is *ALT1*, the alanine aminotransferase ($+0.38$), then *ILV2* and *SER1*. Glutamine's are *KRS1* (-0.41) and *ARG7* (-0.39); arginine's are *TSL1*, *LSP1* and *TPS2* at -0.29 . From the transcriptome, alanine's top reporter is *ALT2* (-0.22), and the top reporter for proline, glutamine, glutamate, lysine and phenylalanine is one gene, *RTG2* of the retrograde response ($+0.16$ to $+0.27$), the pathway that raises glutamate synthesis when mitochondrial function falls. A strain whose proteome carries less of an amino acid's own synthetase or transaminase carries more of the amino acid, which is the direction a pool-and-flux picture predicts and not one a plate artifact would choose.

Table 2: The 19 Muller 2016 amino acids, ordered by the proteome’s score. Ridge r is the out-of-fold Pearson from the whole profile with the permutation 95th percentile in parentheses; the top gene is the single feature with the largest $|r|$ against the metabolite across the shared deletions, with its sign.

SRC: experiments/028-knockout-expression/scripts/expression_metabolic_yko_correlation.py

amino acid	Kemmeren mRNA ridge r (null)	top gene	Messner pro- tein ridge r (null)	top gene	Nadal- Ribelles mRNA ridge r (null)	top gene
proline	0.21 (0.03)	<i>RTG2</i> +0.27	0.61 (0.02)	<i>YHR020W</i> −0.44	0.10 (0.02)	<i>COX4</i> +0.10
alanine	0.31 (0.05)	<i>ALT2</i> −0.22	0.55 (0.02)	<i>ALT1</i> +0.38	0.18 (0.03)	<i>SOD1</i> −0.19
arginine	0.18 (0.03)	<i>YOR366W</i> +0.16	0.52 (0.03)	<i>TSL1</i> −0.29	0.19 (0.04)	<i>TSA1</i> +0.16
glutamine	0.18 (0.05)	<i>RTG2</i> +0.20	0.50 (0.04)	<i>KRS1</i> −0.41	0.09 (0.02)	<i>SOD1</i> +0.12
valine	0.07 (0.05)	<i>OCA1</i> −0.18	0.50 (0.03)	<i>ILV2</i> +0.36	0.01 (0.02)	<i>EFB1</i> +0.08
serine	0.16 (0.03)	<i>DUR12</i> −0.17	0.48 (0.03)	<i>LEU4</i> −0.37	0.20 (0.04)	<i>SOD1</i> +0.21
glycine	−0.01 (0.05)	<i>SRL4</i> −0.11	0.47 (0.02)	<i>SHM2</i> +0.43	0.05 (0.05)	<i>SOD1</i> −0.10
glutamate	0.24 (0.05)	<i>RTG2</i> +0.23	0.46 (0.03)	<i>YHR138C</i> +0.26	0.10 (0.02)	<i>YGL188C</i> +0.09
tyrosine	0.26 (0.06)	<i>GDH2</i> +0.24	0.42 (0.04)	<i>ARO4</i> +0.34	0.12 (0.03)	<i>SOD1</i> −0.20
threonine	0.19 (0.07)	<i>YPS3</i> −0.19	0.41 (0.04)	<i>RNR4</i> −0.20	0.11 (0.05)	<i>TSA1</i> −0.11
histidine	0.23 (0.03)	<i>RDL1</i> −0.15	0.40 (0.04)	<i>ATP4</i> +0.20	0.17 (0.03)	<i>TSA1</i> +0.12
aspartate	0.13 (0.02)	<i>TOP1</i> −0.15	0.40 (0.03)	<i>DIC1</i> +0.22	0.15 (0.05)	<i>AHP1</i> −0.16
methionine	0.25 (0.05)	<i>MCT1</i> −0.25	0.40 (0.03)	<i>ITR1</i> +0.26	0.05 (0.04)	<i>TPI1</i> −0.12
isoleucine	0.21 (0.08)	<i>PDR12</i> +0.17	0.40 (0.03)	<i>GUA1</i> +0.24	0.08 (0.03)	<i>NPL4</i> −0.11
asparagine	0.10 (0.05)	<i>CKA2</i> +0.14	0.38 (0.04)	<i>YHM2</i> −0.21	0.07 (0.02)	<i>TSA1</i> −0.11
leucine	0.09 (0.07)	<i>YPR059C</i> +0.11	0.36 (0.03)	<i>AAT2</i> −0.31	0.12 (0.05)	<i>SOD1</i> +0.15
lysine	0.16 (0.06)	<i>RTG2</i> +0.18	0.32 (0.03)	<i>ARG56</i> +0.16	0.03 (0.03)	<i>ECM16</i> −0.08
phenylalanine	0.12 (0.04)	<i>RTG2</i> −0.16	0.25 (0.02)	<i>TSL1</i> −0.19	0.01 (0.04)	<i>MSK1</i> +0.08
tryptophan	0.10 (0.07)	<i>YHL005C</i> −0.13	0.19 (0.03)	<i>ARG1</i> +0.12	0.04 (0.03)	<i>SHH4</i> −0.09

2.2 The pigment screens read the same way ✗

Betaxanthin, the Cachera 2023 titer proxy, reads 0.43 from the proteome (4,308 shared deletions) and 0.35 from the transcriptome (1,439), against nulls of 0.03. Its top proteins are *COX13* (−0.25), *DLD1* (+0.22) and *ATP7* (−0.22), the respiratory signature the screen’s own top hits carry. The Ozaydin 2013 beta-carotene colony score, an ordinal from −5 to +5 with 45 % of strains at zero, reads 0.16 from protein and 0.12 from mRNA against nulls of 0.04 to 0.05, above chance and far below the amino acids. SRC: experiments/028-knockout-expression/results/expression_metabolic_yko_correlation.json

2.3 Alike transcriptomes, alike amino-acid profiles ✗

Without choosing a metabolite: over the 1.0 million pairs of Kemmeren deletions, the Spearman between profile similarity and amino-acid distance is −0.196 against a permutation 95th percentile of 0.033, and −0.150 on the 282 Cooper deletions with every peak (null 0.077). Strains whose transcriptomes look alike have amino-acid pools that sit close together. From the proteome the same read is −0.046 on Muller (9.7 million pairs, null 0.003) and a null +0.002 on Cooper. So the proteome predicts individual amino acids far better than the transcriptome, yet its whole-profile similarity tracks the amino-acid profile less. Plausible mechanism, unverified: most Messner deletions barely move the proteome, so pairwise profile Pearson among them is noise, while the ridge reads the few proteins that do move.

SRC: experiments/028-knockout-expression/results/expression_metabolic_yko_correlation.json

2.4 Nadal-Ribelles reads the phenotype through one program ✗

The perturb-seq pseudobulk shares 2,113 deletions with Muller and reads a median out-of-fold Pearson of 0.10, 17 of 19 amino acids above nulls of 0.02 to 0.05; serine reads 0.20, arginine 0.19, alanine 0.18, histidine 0.17, and phenylalanine, valine, lysine and tryptophan read at chance. On Cooper it reads a median of 0.01 with 2 of 17 peaks above null, and on the beta-carotene score 0.07 against a null of 0.04. On betaxanthin it reads 0.41 over 2,153 deletions, level with the transcriptome’s 0.35 and the proteome’s 0.43. The structure read is −0.067 on Muller (2.2 million pairs, null 0.034) and −0.059 on Cooper, inside its null of 0.076. SRC: experiments/028-knockout-expression/results/expression_metabolic_yko_correlation.json

Every Nadal read runs through the same reporters. The top single features are the same handful whichever metabolite is asked: *TSA1*, *SOD1*, *AHP1*, *GPX2* and *TRR1* for betaxanthin (−0.33 to −0.27), and *SOD1*, *TSA1*, *AHP1*, *GRE2* for serine, arginine, histidine (+0.11 to +0.21) and alanine (−0.19). These are the oxidative-stress reporters, one transcriptional program. So what the atlas carries about the metabolic phenotype is the position of each deletion on that axis; the amino acids it reads are the ones that axis happens to track, and the ones it cannot

read (phenylalanine, valine, lysine, tryptophan) are the ones the proteome reads through their own enzymes. Plausible mechanism, unverified: the oxidative-stress program is large enough to survive the per-cell counting noise and the batch structure that erase the small, gene-specific responses, which is why the atlas fails to replicate the compendium yet reads a large-effect phenotype as well as the compendium does.

What this says about quality control. Agreement with an earlier compendium is one test of a readout and the metabolic phenotype is another, and the two can disagree. Nadal-Ribelles fails the first (cross-batch r of 0.04, per-deletion agreement with Messner of 0.01) and passes the second on the phenotype that its one program tracks. A readout with no agreement and no metabolic read would have no defense; this one has a narrow defense, and its use is bounded by it: a model trained on it can learn the stress axis and should not be expected to learn the rest.

2.5 The small panels read null, as their size predicts ✗

Zelezniak 2018 shares 93 deletions with Kemmeren and 88 with Messner; da Silveira 2014 shares 121 and 116. A ridge over thousands of features on under 125 samples reads a median out-of-fold Pearson of -0.05 (Zelezniak) and 0.00 to 0.04 (da Silveira), with the da Silveira lipids that clear their null (23 of 147 from mRNA, 34 of 147 from protein) at the rate a 5% threshold produces on its own. These panels say nothing about the question either way.

SRC: `experiments/028-knockout-expression/results/expression_metabolic_yko_correlation.json`

3 The statistics ✗

3.1 Matrices ✗

Each molecular source is a deletion $\times$ feature matrix. Kemmeren is the stored $\log_2$ ratio per reporter, with the $|\log_2| \geq 20$ sentinel cells dropped as absent. Messner is the $\log_2$ of the stored protein quantity over the HIS3 reference profile, duplicate strains of one ORF averaged. Nadal-Ribelles is the recomputed pseudobulk $\log_2$ fold change per deletion (the A statistic of the cross-study recompute), restricted to deletions with at least 50 cells. Each metabolic panel is a deletion $\times$ metabolite matrix whose deletion is the one `kanmx_deletion` perturbation of the record; the pathway cassettes carried by every strain of the pigment screens are the strain background and are ignored. Amino-acid concentrations and metabolite signals are $\log_2$ -transformed, Cooper's zeros dropped first; the betaxanthin corrected fluorescence and the beta-carotene colony score are used as stored. SRC: `experiments/028-knockout-expression/scripts/expression_metabolic_yko_correlation.py`

For one (source, panel) pair, the deletions are those present on both sides, features are kept when measured in at least 95% of those deletions (this is what drops the Messner proteins quantified in few strains), the remainder mean-imputed, and features without variance dropped. A metabolite is scored when at least 40 shared deletions carry a finite value.

3.2 Three reads ✗

Univariate. For each metabolite, the Pearson between the metabolite and every feature across the shared deletions. The report is the largest $|r|$ over features, the number of features past Benjamini-Hochberg FDR 5% on the t test, and the largest $|r|$ under 200 permutations of the deletion labels. The permutation null is the honest reference for a maximum over thousands of features: it sits far above zero, and it widens when the metabolite is heavy tailed, which is why the single-gene read is reported but not relied on.

Multivariate. A ridge from the whole profile to the metabolite, scored out of fold over five folds with the deletions shuffled once by seed; within each training fold the penalty is chosen from $\{10, 10^2, 10^3, 10^4\}$ by an inner three-fold split on the same statistic. The score is the Pearson between the out-of-fold predictions and the metabolite over all shared deletions. The null is the same procedure on 20 permutations of the deletion labels; the folds and their factorizations are fixed by seed, so a permutation reuses them and the null measures the labels, not the split. Each training block is factored once by singular value decomposition, so every target and every penalty is a matrix-vector product.

Structure. A Mantel-style read that needs no choice of metabolite. For every pair of deletions, the Pearson between their two profiles across features (the **profile similarity**) and the Euclidean distance between their z-scored metabolite vectors over deletions with every metabolite present; the statistic is the Spearman between the two over all pairs, and the null is 99 permutations of the deletion labels on the metabolite side. A negative Spearman means that alike profiles go with alike metabolic phenotypes.

4 What the evidence leaves open x

A plate-aware null. Mulleder 2016 and Messner 2023 come from one laboratory on one prototrophic knockout collection, and both were measured plate by plate. A plate effect shared by the two would read as a protein-to-amino-acid signal under a deletion-label permutation, because permuting deletions also permutes plates. The enzyme identities at the top of every strong read argue for biology, but the test is a permutation within plate, which needs the plate assignment of each strain on both sides and has not been run.

The encoder’s cell vector. The proteome carries the amino-acid pools; whether the v14 encoder’s hidden state does is the question that decides the joint head. The read is the same ridge, from the encoder’s cell vector for each deletion to the Mulleder amino acids on the same shared deletions, against the same permutation null. A vector that reads near the proteome’s 0.4 has learned the metabolic phenotype on the way to the proteome; one that reads near zero has not, and the metabolic head would be fitting the encoder rather than reading it.

A joint head on the shared deletions. Some 4,400 deletions carry both a Messner proteome and a Mulleder amino-acid vector. A second node-level head on the amino acids, trained with the proteome head on those deletions, is the direct test of whether the two heads help one another, scored by the proteome’s per-feature Pearson and the amino acids’ out-of-fold Pearson against each head alone.

Spearman for the single-gene read. The maximum $|r|$ over features widens its permutation null under heavy-tailed metabolite values, which is what makes several Kemmeren amino acids show hundreds of features past FDR with a maximum under its null. A rank correlation removes the tails; it is the fix if the single-gene read is ever needed on its own rather than as the illustration it is here.